

Unit 1, Lesson 5: Exploring Rational Exponents



Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.



Learning Targets

- ☐ I can apply the Laws of Exponents to rational exponents.
- ☐ I can convert between expressions involving rational exponents and expressions involving radicals.



1.5.1 Warm-Up: Bell Work

1. Use the Laws of Exponents to write an equivalent expression. Then, find the value of each expression. If necessary, use a calculator to find the value of the equivalent expression.

$(8^{-5})(8^3)$ Equivalent Expression _____ Value _____	$(3^4)^6$ Equivalent Expression _____ Value _____
$\left(\frac{2}{11}\right)^3$ Equivalent Expression _____ Value _____	$\left(\frac{12}{36}\right)^{-4}$ Equivalent Expression _____ Value _____



1.5.2 Exploration: Rational Exponents

1. Fill in the blank with the value of each.

$$16^0 = \underline{\hspace{2cm}}$$

$$16^1 = \underline{\hspace{2cm}}$$

2. Using the values above, guess the value of $16^{\frac{1}{2}}$.
3. Explain the strategy you used.
4. Find the value of $16^{\frac{1}{2}}$ using a calculator.
5. Discuss with your partner what you notice about the relationship between 16 and the value of $16^{\frac{1}{2}}$.

6. Predict the value of $49^{\frac{1}{2}}$. Check using your calculator.
7. Predict the value of $100^{\frac{1}{2}}$. Check using your calculator.
8. Write an expression equivalent to $a^{\frac{1}{2}}$.
9. Check your answer with your partner. If necessary, update your answer.
10. Predict the value of $8^{\frac{1}{3}}$. Check using your calculator.
11. Predict the value of $125^{\frac{1}{3}}$. Check using your calculator.
12. Write an expression equivalent to $a^{\frac{1}{3}}$.
13. Check your answer with your partner. If necessary, update your answer.



1.5.3 Guided Instruction: Why Does That Work?

1. Use the product of powers law, $a^m \cdot a^n = a^{m+n}$, to evaluate the expression.

Expression	Product of Powers	Value
$6^{\frac{1}{2}} \cdot 6^{\frac{1}{2}}$	$6^{\text{---} + \text{---}} = 6^{\text{---}}$	

2. What is the whole number that completes this equation?

$$\sqrt{6} \cdot \sqrt{6} = \text{---}$$

Because $6^{\frac{1}{2}} \cdot 6^{\frac{1}{2}} = \text{---}$ and $\sqrt{6} \cdot \sqrt{6} = \text{---}$, then $6^{\frac{1}{2}}$ is equivalent to --- .

3. Use the product of powers law, $a^m \cdot a^n = a^{m+n}$, to evaluate the expression.

Expression	Product of Powers	Value
$5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}}$	$5^{\text{---} + \text{---} + \text{---}} = 5^{\text{---}}$	

4. What is the whole number that completes the equation?

$$\sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = \text{---}$$

Because $5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} = \text{---}$ and $\sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = \text{---}$, then $5^{\frac{1}{3}}$ is equivalent to --- .

5. Rewrite each exponential expression as a radical expression. Then, evaluate each expression.

Exponential Expression	Radical Expression	Value
$81^{\frac{1}{2}}$		
$8^{\frac{1}{3}}$		
$(-216)^{\frac{1}{3}}$		





1.5.4 Exploration: Other Roots

Rewriting exponential expressions as radical expressions does not only apply to square roots and cube roots. Other roots apply these same principals.

Complete the table to explore other rational exponents and roots.

Rational Exponent	Radical Expression	Value
$81^{\frac{1}{4}}$		
	$\sqrt[4]{256}$	
	$\sqrt[5]{-32}$	
$\left(1,000,000,000^{\frac{1}{9}}\right)$		



1.5.5 Guided Instruction: Exponent Laws With Rational Exponents

- Kalini's teacher asked her to rewrite $\left((100)^{\frac{1}{2}}\right)^3$ as an equivalent exponential expression. Kalini says that she can use a law of exponents to multiply the powers.
 - Which law of exponents is Kalini using?
 - Using the laws of exponents, what is an equivalent exponential expression for $\left((100)^{\frac{1}{2}}\right)^3$?
- The expression $27^{\frac{1}{3}} \cdot 27^{\frac{5}{3}}$ can also be rewritten using the laws of exponents.
 - Rewrite the expression.
$$27^{\frac{1}{3}} \cdot 27^{\frac{5}{3}} =$$
 - Which exponent law was used?
 - Check your answer with your partner.
 - Evaluate the exponential expression.

3. Maeko explained his work for rewriting the following expression.

Explanation	$16^{-\frac{8}{2}} \cdot 16^{\frac{1}{2}}$
I first used the product of powers to rewrite the expression.	$16^{-\frac{8}{2} \cdot \frac{1}{2}} = 16^{-\frac{8}{4}}$
Next, I simplified the fractional exponent.	4^{-2}
Finally, I rewrote the expression using the negative exponent law.	$\frac{1}{16}$

Check Maeko's work and explain any errors he may have made.



1.5.6 Try It: Writing Rational Exponents

Rewrite each expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$5^{\frac{26}{7}} \cdot 5^{\frac{9}{7}}$		
$\frac{23^{\frac{8}{3}}}{23^{\frac{2}{3}}}$		
$125^{\frac{7}{15}} \cdot 125^{-\frac{2}{15}}$		
$169^{-\frac{5}{22}} \cdot 169^{\frac{25}{22}} \cdot 169^{-\frac{31}{22}}$		



1.5.7 Wrap-Up: Ticket Out the Door

1. Find the value of the variable.

a. $\sqrt{14} = 14^{\frac{1}{k}}$

$k = \underline{\hspace{2cm}}$

b. $50^{\frac{1}{3}} = \sqrt[n]{50}$

$n = \underline{\hspace{2cm}}$

2. Complete the table by writing an equivalent expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$6^{-\frac{1}{6}} \cdot 6^{\frac{19}{6}}$		
$\frac{7^{\frac{1}{2}}}{7^{\frac{5}{2}}}$		

Unit 1, Lesson 6: Writing Equivalent Expressions Using Rational Exponents



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